

A PROJECT TITLE
ON
Environmental Sensor Data Visualization Using PCA & UMAP

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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DEPARTMENT OF Computer Science Engineering

Name of Title: **Environmental Sensor Data Visualization Using PCA & UMAP**

1. Abstract:

Environmental monitoring plays a vital role in understanding changes in weather conditions, air quality, and pollution levels. Modern environmental sensors continuously generate large amounts of data, including temperature, humidity, air pressure, and air quality measurements. Since this data contains multiple features, it becomes difficult to analyse and visualize using traditional methods. This project, "Environmental Sensor Data Visualization Using PCA and UMAP," focuses on simplifying complex environmental sensor data through dimensionality reduction techniques.

The project uses Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) to reduce high-dimensional sensor data into a two-dimensional representation while preserving important information. PCA identifies the directions of maximum variance in the dataset, whereas UMAP effectively preserves the relationships between similar data points, making hidden patterns and clusters easier to observe. Before applying these techniques, the sensor data is pre-processed by handling missing values, removing inconsistencies, and scaling the features to improve analysis accuracy.

The transformed data is visualized using scatter plots, enabling users to compare the performance of PCA and UMAP in representing environmental conditions. The implementation is carried out using Python with libraries such as Pandas, NumPy, Scikit-learn, UMAP-learn, and Matplotlib. The project provides an effective way to explore large environmental datasets, identify trends, detect unusual observations, and support better decision-making. It demonstrates how dimensionality reduction and visualization techniques can make complex environmental sensor data easier to understand and useful for applications such as pollution monitoring, climate studies, smart agriculture, and smart city management.

2. INTRODUCTION:

Environmental monitoring has become increasingly important for understanding weather conditions, air quality, and pollution levels. Modern environmental sensors continuously collect data such as temperature, humidity, air pressure, and air quality, resulting in large and complex datasets. Analysing such high-dimensional data using traditional methods can be difficult and time-consuming.

This project, "Environmental Sensor Data Visualization Using PCA and UMAP," aims to simplify environmental sensor data by applying two dimensionality reduction techniques: **Principal Component Analysis (PCA)** and **Uniform Manifold Approximation and Projection (UMAP)**. These techniques

reduce the number of data dimensions while preserving important information, making it easier to visualize patterns, trends, and clusters.

The project includes data preprocessing, dimensionality reduction, and visualization using scatter plots. It is implemented using Python with libraries such as Pandas, NumPy, Scikit-learn, UMAP-learn, and Matplotlib. The proposed system helps users better understand environmental data and supports applications such as pollution monitoring, climate analysis, smart agriculture, and smart city development.

3. SOFTWARE REQUIREMENTS:

The following software is required to develop the project:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.x
- IDE/Editor: Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- Libraries:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn (sklearn)
 - UMAP (umap-learn)
- Dataset: Environmental Sensor Dataset (CSV format)



Signature of the Guide



Course Instructor